Real data form clusters, dense conglomerates gather many points close to each other, alternated to sparse areas with few points. With a regularly spaced grid, the risk is to have many empty tiles and a few tiles gathering hundreds or thousands of points, hence defying the purpose of this approach. We need something different, more flexible.

8.2.3 Simplify Things to Get a Hint

The solution to this problem seems, indeed, elusive. Sometimes, in these cases, it helps trying to solve a simplified version of the problem, and then come up with a more general solution that works for our initial problem.

Suppose, for instance, that we were able to restrict us to a 1-D search space: say that we need to serve only customers on a single road, that extends for miles and miles, and all our warehouses are also placed along this same road.

To simplify things even further, suppose that the road is perfectly straight, and that the total distance we cover is short enough that we don’t have to worry about Earth’s surface being curve, latitude, longitude etc… Basically, we assume an approximation with a 1-D segment is close enough to our space, and we can use Euclidean distance in 1-D as a proxy for the real distance between cities.